OGC® DOCUMENT: 23-004

Open Geospatial Consortium

OGC API - CONNECTED SYSTEMS - PART 4: SAMPLING FEATURES

STANDARD
Implementation

CANDIDATE SWG DRAFT

Version: 1.0
Submission Date: XXX
Approval Date: XXX
Publication Date:
Editor: Alexandre Robin

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Note

CONTENTS

I ABSTRACT

This working draft organizes the current Part 4 sampling-features source already present in the repository as a standalone draft entrypoint. The integrated source currently provides a references clause together with the requirements classes for sampling feature types and their supporting requirement files and figures. Additional standalone-draft elements, including fuller conformance packaging and annex treatment, remain under editorial cleanup on this branch.

II KEYWORDS

The following are keywords to be used by search engines and document catalogues.

ogcdoc, OGC document, OpenAPI, REST, API, connected system, sampling feature, specimen, statistical sample, spatial sampling, parametric sampling

III PREFACE

The OGC API — Connected Systems Standard is part of the suite of OGC API Standards.

This working-draft branch packages the current Part 4 sampling-features source material as a standalone draft for review while broader standalone-draft cleanup remains in progress.

In the context of this document, the following phrases are used:

- “this Standard” refers to “OGC API — Connected Systems — Part 4: Sampling Features”.
- “CS API” or “CS API Standard” refers to the multi-part OGC API — Connected Systems Standard.

IV SECURITY CONSIDERATIONS

No security considerations have been made for this document.

V SUBMITTERS

This working-draft branch currently preserves editor metadata in the root document header. A complete Part 4 submitters table has not yet been promoted into the standalone draft from source material available in this branch.

1

SCOPE

1 SCOPE

This working draft packages the existing Part 4 material for sampling feature types as a standalone part of the CS API Standard.

The currently integrated source specifies requirements for several types of `Sampling Features` that can be used to document different sampling strategies. These include the sampling feature type requirements carried by the current `api/part4` source tree and the supporting figures and requirement-class includes already present in the repository.

A fuller standalone Part 4 scope statement has not yet been promoted from source material available in this branch. Until that editorial cleanup is completed, this clause should be read as a bounded description of the source currently integrated into the working draft.

2

CONFORMANCE

2 CONFORMANCE

This working-draft branch preserves the extracted Part 4 requirement-class source and its supporting references as currently available in the repository.

The standalone conformance packaging for Part 4 is not yet complete in this branch. Requirements and conformance-linkage cleanup remains an explicit follow-on task, and unresolved linkage warnings should be treated as editorial cleanup items rather than as silently resolved structure.

3

NORMATIVE REFERENCES

3 NORMATIVE REFERENCES

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

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Nottingham M: IETF RFC 8288, *Web Linking*. RFC Publisher (2017). https://www.rfc-editor.org/info/rfc8288.

JSON Schema Validation: A Vocabulary for Structural Validation of JSON, Version 2020-12, https://json-schema.org/draft/2020-12/json-schema-validation.html

4

TERMS AND DEFINITIONS

4 TERMS AND DEFINITIONS

No terms and definitions are listed in this document.

All terms and definitions used by the integrated Part 4 source remain as defined by the referenced standards and related parts of the CS API Standard.

A standalone Part 4 terms-and-definitions clause has not yet been promoted into this working draft from source material available in this branch. Until that work is completed, reviewers should read the integrated sampling-feature-types clause together with the referenced standards and related CS API parts it cites.

5

REQUIREMENTS CLASSES FOR SAMPLING FEATURE TYPES

5 REQUIREMENTS CLASSES FOR SAMPLING FEATURE TYPES

5.1. Overview

This section specifies requirements for several types of `Sampling Features` that can be used to document different sampling strategies. More sampling feature types can be defined by extensions.

See the Encoding Requirements Classes for examples of `Sampling Feature` resource representations.

5.2. Requirements Class “Spatial Sampling Features”

5.2.1. Overview

REQUIREMENTS CLASS 1	
IDENTIFIER	`/req/sampling-spatial`
TARGET TYPE	Web API
PREREQUISITE	/req/sampling-features
INDIRECT PREREQUISITE	OGC 10-004r3 — Clause 9: Spatial Sampling Features
NORMATIVE STATEMENTS	Requirement 1: `/req/sampling-spatial/type` Requirement 2: `/req/sampling-spatial/om` Requirement 3: `/req/sampling-spatial/shapes` Requirement 4: `/req/sampling-spatial/location` Requirement 5: `/req/sampling-spatial/location-time`

`Spatial Sampling Features` are generic sampling features with a sampling location or geometry, in one, two or three dimensions. They are used when no physical sample is collected and observations are made directly on a spatial subset of the sampled feature (e.g. an in-situ measurement station, a profile through a gaseous or liquid medium, an image footprint, etc.). The original model for these feature types is defined as part of Observations and Measurements v2.0.

NOTE 1: The `Spatial Sampling Features` introduced in this section are best used for modeling static or slowly moving features. Although it is possible to handle dynamic use cases using these classes, it requires posting a new feature to the server everytime the location of the feature changes. This can be very inefficient for highly mobile sampling features such as the ones attached to moving platforms or rotating sensors (e.g PTZ cams). Parametric sampling features defined in Clause 5.6 are better suited for these use cases since they can be positioned relatively to a moving platform.

NOTE 2: However, even for mobile cases, `Spatial Sampling Features` can be useful if capturing a different feature for each new observation is desired. For instance, this is the case for space-borne earth observation where the geographic footprint of each acquisition is typically provided along with the collected imagery.

Below are a few examples for each sub-type of `Spatial Sampling Feature`:

Sampling Point Examples

- A sampling location at a river monitoring station.
- A sampling location at the bottom of a well.

Sampling Curve Examples

- A sampling path along a river.
- A sampling profile below an ocean monitoring buoy.

Sampling Surface Examples

- A satellite image footprint
- The geospatial extent covered by a weather forecast model.

5.2.2. Feature Type

REQUIREMENT 1	
IDENTIFIER	`/req/sampling-spatial/type`
INCLUDED IN	Requirements class 1: `/req/sampling-spatial`
STATEMENT	A `Spatial Sampling Feature` resource SHALL use one of the URIs or CURIEs provided in Table 1 as the value of the `Feature Type` attribute.

REQUIREMENT 1	
	If using a URI or CURIE is not practical, the encoding SHALL explicitely define the mapping between the values used and these URIs.

Table 1 — Sampling Spatial Feature Types

SYSTEM TYPE	URI	CURIE	EXAMPLE USAGE
Sampling Point	http://www.opengis.net/def/samplingFeatureType/OGC-OM/2.0/SF_SamplingPoint	om:SamplingPoint	Observing station location
Sampling Curve	http://www.opengis.net/def/samplingFeatureType/OGC-OM/2.0/SF_SamplingCurve	om:Sampling Curve	Profile, Path
Sampling Surface	http://www.opengis.net/def/samplingFeatureType/OGC-OM/2.0/SF_SamplingSurface	om:Sampling Surface	Image footprint
Sampling Solid	http://www.opengis.net/def/samplingFeatureType/OGC-OM/2.0/SF_SamplingSolid	om:SamplingSolid	Radar lobe

5.2.3. Attributes

REQUIREMENT 2	
IDENTIFIER	`/req/sampling-spatial/om`
INCLUDED IN	Requirements class 1: `/req/sampling-spatial`
STATEMENT	A `Spatial Sampling Feature` resource SHALL implement the SF_SpatialSamplingFeature class defined in OGC 10-004r3.

Support for attributes listed in the following table is required:

Table 2 — Common Sampling Feature Attributes

NAME	DEFINITION	DATA TYPE	USAGE
`shape`	The location or geometry of the spatial sampling feature.	Geometry	Required

REQUIREMENT 3	
IDENTIFIER	`/req/sampling-spatial/shapes`

REQUIREMENT 3	
INCLUDED IN	Requirements class 1: `/req/sampling-spatial`
A	An implementation of this feature type SHALL support the sub types `SamplingPoint`, `SamplingCurve`, and `SamplingSurface`. Sub type `SamplingSolid` is not required.
B	An implementation of this feature type SHALL support `Point`, `Linestring` and `Polygon` geometries (to be used respectively with `SamplingPoint`, `SamplingCurve`, and `SamplingSurface` features).

5.2.3.1. Location

Spatial Sampling Features always include a shape (or geometry). For mobile sampling features, this shape must be the latest known location/geometry of the feature.

REQUIREMENT 4	
IDENTIFIER	`/req/sampling-spatial/location`
INCLUDED IN	Requirements class 1: `/req/sampling-spatial`
STATEMENT	A `Sampling Feature` resource SHALL include the latest known location/geometry of the feature.

If the parameter `datetime` is used, the geometry must be the geometry of the feature known at that time.

REQUIREMENT 5	
IDENTIFIER	`/req/sampling-spatial/location-time`
INCLUDED IN	Requirements class 1: `/req/sampling-spatial`
A	If the implementation supports mobile sampling features, the location/geometry SHALL be reported according to the `datetime` request parameter.
B	When the `datetime` parameter is ommitted, the server SHALL report the latest known geometry of the feature.
C	When the `datetime` parameter is set to a time instant, the server SHALL report the latest geometry of the feature that was known at this point in time.
D	When the `datetime` parameter is set to a time period, the server SHALL report the latest location of the feature that was known at the end of the specified time period.

5.3. Requirements Class “Specimen Features”

5.3.1. Overview

REQUIREMENTS CLASS 2	
IDENTIFIER	`/req/sampling-specimen`
TARGET TYPE	Web API
PREREQUISITE	/req/sampling-features
INDIRECT PREREQUISITE	OGC 10-004r3 — Clause 10: Specimens
NORMATIVE STATEMENTS	Requirement 6: `/req/sampling-specimen/type` Requirement 7: `/req/sampling-specimen/om`

A `Specimen Feature` respresents a physical sample, obtained for observation(s) carried out ex-situ, sometimes in a laboratory.

NOTE: A generalization of this class is referred to as *Material Sample* in Clause 13.4 of Observations, Measurements and Samples, Version 3.0.

Examples of Specimens

- Soil sample
- Water sample
- Rock core
- Blood sample
- Tissue sample
- Vegetation sample
- Food sample

5.3.2. Feature Type

REQUIREMENT 6	
IDENTIFIER	`/req/sampling-specimen/type`
INCLUDED IN	Requirements class 2: `/req/sampling-specimen`
STATEMENT	The `Feature Type` attribute of a `Specimen Feature` resource SHALL be set to: • The URI http://www.opengis.net/def/samplingFeatureType/OGC-OM/2.0/SF_Specimen, or • The CURIE `om:Specimen`. If using a URI or CURIE is not practical, the encoding SHALL unambiguously map one of the possible values used to encode feature types to this URI.

5.3.3. Attributes

REQUIREMENT 7	
IDENTIFIER	`/req/sampling-specimen/om`
INCLUDED IN	Requirements class 2: `/req/sampling-specimen`
STATEMENT	A `Specimen Feature` resource SHALL implement the SF_Specimen class defined in OGC 10-004r3.

The attributes of this class are recalled below:

Table 3 — Specimen Attributes

NAME	DEFINITION	DATA TYPE	USAGE
`materialClass`	Basic classification of the material type of the specimen. (e.g. soil, water, rock, aqueous, liquid, tissue, vegetation, food)	QName	Required
`samplingTime`	Time when the specimen was retrieved from the sampled feature.	TimeInstant	Required
`samplingLocation`	The location from where the sample was obtained.	Geometry	Optional
`samplingMethod`	The location from where the sample was obtained.	QName	Optional
`storageLocation`	The location where the sample is stored.	Named Location	Optional
`specimenType`	The description of the basic form of the specimen. (e.g. polished section; core; pulp; solution)	QName	Optional

NAME	DEFINITION	DATA TYPE	USAGE
`size`	The physical extent of the sample. (e.g. length, mass, volume, etc. as appropriate for the instance and its material type)	Measure	Optional

5.4. Requirements Class “Statistical Sample”

5.4.1. Overview

REQUIREMENTS CLASS 3	
IDENTIFIER	`/req/statistical-sample`
TARGET TYPE	Web API
PREREQUISITE	/req/sampling-features
INDIRECT PREREQUISITE	http://www.opengis.net/spec/om/3.0/req/sam-basic/StatisticalSample
NORMATIVE STATEMENTS	Requirement 8: `/req/statistical-sample/type` Requirement 9: `/req/statistical-sample/attributes`

A `StatisticalSample` is a statistical subset of a feature-of-interest, defined for the purpose of creating observation(s).

This type of sampling feature is defined in Clause 13.5 of Observations, Measurements and Samples, Version 3.0.

Examples of Statistical Samples

- The male or female subset of a population
- 200 persons from age group 19-25
- A random subset of the oak trees in a forest
- A random subset of the manufactured goods coming out of a production line

5.4.2. Feature Type

REQUIREMENT 8	
IDENTIFIER	`/req/statistical-sample/type`
INCLUDED IN	Requirements class 3: `/req/statistical-sample`
STATEMENT	The `Feature Type` attribute of a `Statistical Sample` resource SHALL be set to: • The URI http://www.w3.org/ns/sosa/oms/StatisticalSample, or • The CURIE `oms:StatisticalSample`. If using a URI or CURIE is not practical, the encoding SHALL unambiguously map one of the possible values used to encode feature types to this URI.

5.4.3. Attributes

REQUIREMENT 9	
IDENTIFIER	`/req/statistical-sample/attributes`
INCLUDED IN	Requirements class 3: `/req/statistical-sample`
STATEMENT	A `Statistical Sample` feature resource SHALL implement the `StatisticalSample` class defined in OGC 20-082r4.

The attributes of this class are recalled below:

Table 4 — Material Sample Attributes

NAME	DEFINITION	DATA TYPE	USAGE
`classification`	The criterion by which the subset was defined.	CodeListValue	Optional

5.5. Requirements Class “Feature Parts”

5.5.1. Overview

REQUIREMENTS CLASS 4	
IDENTIFIER	`/req/feature-part`
TARGET TYPE	Web API
PREREQUISITE	/req/sampling-features
NORMATIVE STATEMENTS	Requirement 10: `/req/feature-part/type` Requirement 11: `/req/feature-part/attributes`

`Feature Parts` are used to clearly identify the part of a feature that is being observed or controlled within a larger feature. They are useful to provide more details about the sampling or control point when the part is not itself already described as a feature object.

NOTE 1: Feature parts are particularly useful when the feature of interest is a `System` resource. They allow to precisely identify what part of the system is being observed or controlled (e.g. the wing of an aircraft, the engine of a car, etc.)

NOTE 2: Since this API does not allow linking an observation to the ultimate feature of interest directly, `Feature Parts` can also be used as a proxy to refer to an existing feature of interest as a whole. This is especially useful when referring to ultimate features of interest that are hosted on third party servers.

Examples of Feature Parts

- The CPU in a computer or robot
- The center of mass of a UAV
- The antenna of a GPS receiver
- The vent of an A/C system
- A room or door in a building
- The entire vehicle (proxy to the system feature describing the vehicle as a `Platform`)

5.5.2. Feature Type

REQUIREMENT 10	
IDENTIFIER	`/req/feature-part/type`
INCLUDED IN	Requirements class 4: `/req/feature-part`
STATEMENT	The `Feature Type` attribute of a `Feature Part` resource SHALL be set to:

REQUIREMENT 10	
	- The URI http://www.opengis.net/def/samplingFeatureType/x-CS-TBD/1.0/FeaturePart, or - The CURIE `cs:FeaturePart`. If using a URI or CURIE is not practical, the encoding SHALL unambiguously map one of the possible values used to encode feature types to this URI.

5.5.3. Attributes

REQUIREMENT 11	
IDENTIFIER	`/req/feature-part/attributes`
INCLUDED IN	Requirements class 4: `/req/feature-part`
A	A `Feature Part` sampling feature SHALL include all mandatory attributes listed in Table 5.
B	The value assigned to these attributes SHALL conform to the specified definition and data type.
C	All attributes marked as `Required` SHALL be present on an instance of the class.

Table 5 — Feature Part Attributes

NAME	DEFINITION	DATA TYPE	USAGE
`definition`	Points to an ontology/dictionary entry that defines the nature of the part.	URI	Optional
`partId`	Identifier of the part if there is one.	string	Optional

NOTE: The `partId` can refer to the manufacturer nomenclature or to the object ID in a 3D model for example.

5.6. Requirements Class “Parametric Sampling Features”

5.6.1. Overview

REQUIREMENTS CLASS 5	
IDENTIFIER	`/req/sampling-parametric`
TARGET TYPE	Web API
PREREQUISITES	/req/sampling-features http://www.opengis.net/spec/GeoPose/1.0/req/basic-ypr http://www.opengis.net/spec/GeoPose/1.0/req/basic-quaternion http://www.opengis.net/spec/SML/2.1/req/relative-ypr http://www.opengis.net/spec/SML/2.1/req/relative-quaternion
NORMATIVE STATEMENTS	Requirement 12: `/req/sampling-parametric/attributes` Requirement 13: `/req/sampling-parametric/pose`

Parametric sampling features are special kinds of spatial sampling features whose geometry is defined implicitely by a certain number of parameters, rather than by vertex coordinates. These parameters include extrinsic parameters defining the pose of the feature relative to a coordinate frame of reference (outer frame), and intrinsic parameters defining the shape of the feature in its own coordinate frame (inner frame).

One advantage of parametric sampling features is their capability for relative positioning via the `pose` attribute. Since they can be positionned relatively to a moving object, their parameters don’t need to be updated frequently when the object moves. Rather, the sampling feature would “track” the object automatically.

All parametric sampling features share some requirements that are detailed below.

5.6.2. Attributes

REQUIREMENT 12	
IDENTIFIER	`/req/sampling-parametric/attributes`
INCLUDED IN	Requirements class 5: `/req/sampling-parametric`
A	A `Parametric Sampling Feature` resource SHALL support the attributes listed in Table 6.

REQUIREMENT 12	
B	The value assigned to these attributes SHALL conform to the specified definition and data type.
C	All attributes marked as `Required` SHALL be present on an instance of the class.

Table 6 — Parametric Sampling Feature Attributes

NAME	DEFINITION	DATA TYPE	USAGE
`pose`	The pose of the sampling feature, relative to the base reference frame.	Pose	Optional

5.6.3. Pose

Every parametric sampling feature has a `pose` attribute that takes an instance of a `Pose` object based on the SensorML pose model, itself derived from the GeoPose model.

The `pose` property allows the specification of the feature location and orientation relatively to a frame of reference. More specifically, it provides the pose of the inner frame relative to the outer frame (the frame of reference).

The allowed values for the `pose` attribute are "GeoPose Basic" instances or "SensorML Relative Pose" instances.

REQUIREMENT 13	
IDENTIFIER	`/req/sampling-parametric/pose`
INCLUDED IN	Requirements class 5: `/req/sampling-parametric`
A	The value of the `pose` attribute SHALL be one of the following: • A GeoPose Basic-YPR instance • A GeoPose Basic-Quaternion instance • A SensorML Relative-YPR instance • A SensorML Relative-Quaternion instance

When using a GeoPose `Basic-YPR` or `Basic-Quaternion` instance, the position is always a geographic location expressed in the EPSG:4979 (WGS 84, lat/lon/alt) coordinate reference system, and the orientation is relative to the ENU LTP frame located at the geographic location. Thus, this is only suited for static sampling features.

For other use cases (e.g. mobile), a SensorML `Relative-YPR` or `Relative-Quaternion` pose instance is required. In this case, the coordinate frame of reference used to express the pose can be any cartesian coordinate frame, but the most frequently used ones are:

- A local tangent plane (LTP) coordinate frame such as NED or ENU
- An engineering coordinate frame attached to a system (fixed or mobile, e.g. a UAV platform)
- An engineering coordinate frame attached to another feature of interest (e.g. building)

When using the SensorML `Relative-YPR` or `Relative-Quaternion` instance with an LTP frame, the location of the frame is provided by the feature location attribute that takes a point geometry, and the position in the pose is an offset from that location, expressed in the LTP frame (before the rotation).

When an LTP coordinate frame is used, the location property of the sampling feature is used to indicate the point in space where the LTP is located (typically a geographic location).

5.6.4. Location

The location can be used as the default point location for clients that don't understand the extended properties.

5.7. Requirements Class "Relative Sampling Point"

5.7.1. Overview

REQUIREMENTS CLASS 6	
IDENTIFIER	`/req/relative-sampling-point`
TARGET TYPE	Web API
PREREQUISITES	/req/sampling-features Requirements class 5: `/req/sampling-parametric`
NORMATIVE STATEMENTS	Requirement 14: `/req/relative-sampling-point/type` Requirement 15: `/req/relative-sampling-point/pose`

The `Relative Sampling Point` class allows to define a sampling point relatively to a cartesian reference frame.

Relative Sampling Points Examples

- X/Y/Z position relative to a surface location, expressed in the ENU local tangent frame
- X/Y/Z position relative to a moving platform (e.g. under water sampling point on a USV)

The orientation part of the pose is not used in the `Relative Sampling Point` class and it is illegal to provide it.

5.7.2. Feature Type

REQUIREMENT 14	
IDENTIFIER	`/req/relative-sampling-point/type`
INCLUDED IN	Requirements class 6: `/req/relative-sampling-point`
STATEMENT	The `Feature Type` attribute of a `Relative Sampling Point` resource SHALL be set to: • The URI http://www.opengis.net/def/samplingFeatureType/x-CS-TBD/1.0/RelativeSamplingPoint, or • The CURIE `cs:RelativeSamplingPoint`. If using a URI or CURIE is not practical, the encoding SHALL unambiguously map one of the possible values used to encode feature types to this URI.

5.7.3. Attributes

This Standard does not define any additional attributes for the `Relative Sampling Point` feature type. However, the following requirements must be fullfilled:

REQUIREMENT 15	
IDENTIFIER	`/req/relative-sampling-point/pose`
INCLUDED IN	Requirements class 6: `/req/relative-sampling-point`
STATEMENT	The `pose` attribute SHALL contain the . It SHALL not contain any orientation information.

5.8. Requirements Class “Sampling Sphere”

5.8.1. Overview

REQUIREMENTS CLASS 7	
IDENTIFIER	`/req/sampling-sphere`
TARGET TYPE	Web API
PREREQUISITES	/req/sampling-features Requirements class 5: `/req/sampling-parametric`
NORMATIVE STATEMENTS	Requirement 16: `/req/sampling-sphere/type` Requirement 17: `/req/sampling-sphere/attributes`

5.8.2. Feature Type

REQUIREMENT 16	
IDENTIFIER	`/req/sampling-sphere/type`
INCLUDED IN	Requirements class 7: `/req/sampling-sphere`
STATEMENT	The `Feature Type` attribute of a `Sampling Sphere` resource SHALL be set to: • The URI http://www.opengis.net/def/samplingFeatureType/x-CS-TBD/1.0/SamplingSphere, or • The CURIE `cs:SamplingSphere`. If using a URI or CURIE is not practical, the encoding SHALL unambiguously map one of the possible values used to encode feature types to this URI.

5.8.3. Attributes

REQUIREMENT 17	
IDENTIFIER	`/req/sampling-sphere/attributes`

REQUIREMENT 17	
INCLUDED IN	Requirements class 7: `/req/sampling-sphere`
A	A `Sampling Sphere` feature SHALL support the attributes listed in Table 7.
B	The value assigned to these attributes SHALL conform to the specified definition and data type.
C	All attributes marked as `Required` SHALL be present on an instance of the class.

Table 7 — Sampling Sphere Attributes

NAME	DEFINITION	DATA TYPE	USAGE
`radius`	Radius of the sphere in meters.	Number	Required

5.9. Requirements Class “Sampling Profile”

5.9.1. Overview

REQUIREMENTS CLASS 8	
IDENTIFIER	`/req/sampling-profile`
TARGET TYPE	Web API
PREREQUISITES	/req/sampling-features Requirements class 5: `/req/sampling-parametric`
NORMATIVE STATEMENTS	Requirement 18: `/req/sampling-profile/type` Requirement 19: `/req/sampling-profile/attributes`

A `Sampling Profile` is used when measurements are taken at various distances along a straight line, at regular or irregular intervals.

The location of the profile “bins” is defined relatively to the origin of the profile, which makes it suitable for profiles attached to moving platforms.

NOTE: Although a `SamplingCurve` feature can be used to model a profile, it requires the specification of the 3D location of each profile bin, separately. This makes it quite inefficient when ingesting data for mobile profilers. But note that it can still be used to read the data back as the `SamplingCurve` feature can be reconstructed dynamically knowing the platform location and sampling profile parameters.

Examples of Sampling Profiles

- Sampling at various depth within a borehole
- Sampling at various depth by an acoustic doppler current profiler (ADCP) attached to an ocean buoy
- Sampling at various distances along the radial of a weather Doppler radar
- Sampling at various heights above an atmospheric sounder

5.9.2. Feature Type

REQUIREMENT 18	
IDENTIFIER	`/req/sampling-profile/type`
INCLUDED IN	Requirements class 8: `/req/sampling-profile`
STATEMENT	The `Feature Type` attribute of a `Sampling Profile` resource SHALL be set to: • The URI http://www.opengis.net/def/samplingFeatureType/x-CS-TBD/1.0/SamplingProfile, or • The CURIE `cs:SamplingProfile`. If using a URI or CURIE is not practical, the encoding SHALL unambiguously map one of the possible values used to encode feature types to this URI.

5.9.3. Attributes

REQUIREMENT 19	
IDENTIFIER	`/req/sampling-profile/attributes`
INCLUDED IN	Requirements class 8: `/req/sampling-profile`
A	A `Sampling Profile` feature SHALL support the attributes listed in Table 8.

REQUIREMENT 19	
B	The value assigned to these attributes SHALL conform to the specified definition and data type.
C	All attributes marked as `Required` SHALL be present on an instance of the class.

Table 8 — Sampling Profile Attributes

NAME	DEFINITION	DATA TYPE	USAGE
`pose`	The position and orientation of the profile, relative to the outer reference frame	GeoPose	Required
`profileAxis`	Axis along which the profile bins are located, in the local frame defined by the pose information (the axis is pointing in the direction of increasing range)	Enum ['X', 'Y', 'Z', '-X', '-Y', '-Z']	Required
`binDistances`	Distances of bins along the profile, in meters	Array of Numbers	Required
`spreadAngle`	Beam spread angle, in degrees (defaults to 0)	Number	Optional

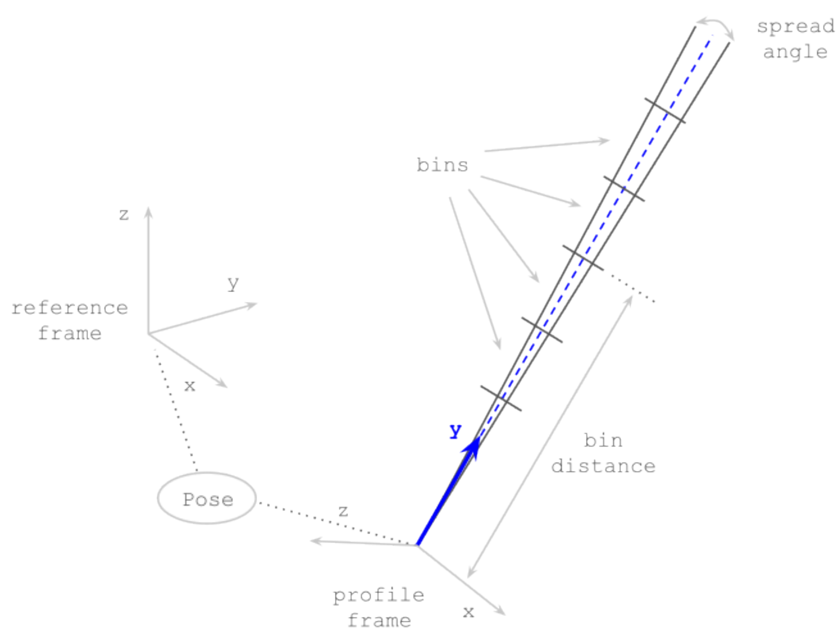

Figure 1 — Sampling Profile (along Y)

5.10. Requirements Class “Viewing Frustum”

5.10.1. Overview

REQUIREMENTS CLASS 9	
IDENTIFIER	`/req/viewing-frustum`
TARGET TYPE	Web API
PREREQUISITES	/req/sampling-features Requirements class 5: `/req/sampling-parametric`
NORMATIVE STATEMENTS	Requirement 20: `/req/viewing-frustum/type` Requirement 21: `/req/viewing-frustum/attributes`

`Viewing Frustums` are used to model the sampling volume of remote sensors, such as (pinhole) cameras and other kinds of imagers. For an optical sensor, the viewing frustum contains the extent of the observable world that is visible within its field of view.

NOTE: The real viewing volume of an optical imager has a more complex shape due to distortion but a frustum is usually enough to get a good approximation of the area imaged by such a sensor.

Examples of Viewing Frustums

- The viewing frustum of a video camera or imager
- The viewing frustum of the human eye
- The viewing frustum of an IR motion detector
- The frustum of a light projection system

5.10.2. Feature Type

REQUIREMENT 20	
IDENTIFIER	`/req/viewing-frustum/type`
INCLUDED IN	Requirements class 9: `/req/viewing-frustum`

REQUIREMENT 20

STATEMENT	The `Feature Type` attribute of a `Viewing Frustum` resource SHALL be set to: • The URI http://www.opengis.net/def/samplingFeatureType/x-CS-TBD/1.0/ViewingFrustum, or • The CURIE `cs:ViewingFrustum`. If using a URI or CURIE is not practical, the encoding SHALL unambiguously map one of the possible values used to encode feature types to this URI.

5.10.3. Attributes

REQUIREMENT 21

IDENTIFIER	`/req/viewing-frustum/attributes`
INCLUDED IN	Requirements class 9: `/req/viewing-frustum`
A	A `Viewing Frustum` feature SHALL support the attributes listed in Table 9.
B	The value assigned to these attributes SHALL conform to the specified definition and data type.
C	All attributes marked as `Required` SHALL be present on an instance of the class.

Table 9 — Viewing Frustum Attributes

NAME	DEFINITION	DATA TYPE	USAGE
`pose`	The position and orientation of the frustum, relative to the outer reference frame.	GeoPose	Required
`frustumAxis`	Longitudinal axis of the frustum, expressed in the inner frame defined by the pose information (the axis is pointing from the frustum apex toward its base).	Enum ['X', 'Y', 'Z', '-X', '-Y', '-Z']	Required
`upAxis`	Lateral axis of the frustum, pointing in the image up direction, expressed in the inner frame defined by the pose information.	Enum ['X', 'Y', 'Z', '-X', '-Y', '-Z']	Required
`fov`	Field of view, in degrees. If the aspect ratio is also provided, this is the horizontal field of view of a pyramidal frustum (orthogonal to the up direction), otherwise the frustum is considered conical.	Number	Required
`aspectRatio`	Aspect ratio of the frustum base rectangle, that is the ratio of its width to its height (unitless).	Number	Optional

NAME	DEFINITION	DATA TYPE	USAGE
`length`	Length of the frustum, in meters.	Number	Optional

The shape and parameters of a pyramidal frustum are illustrated below:

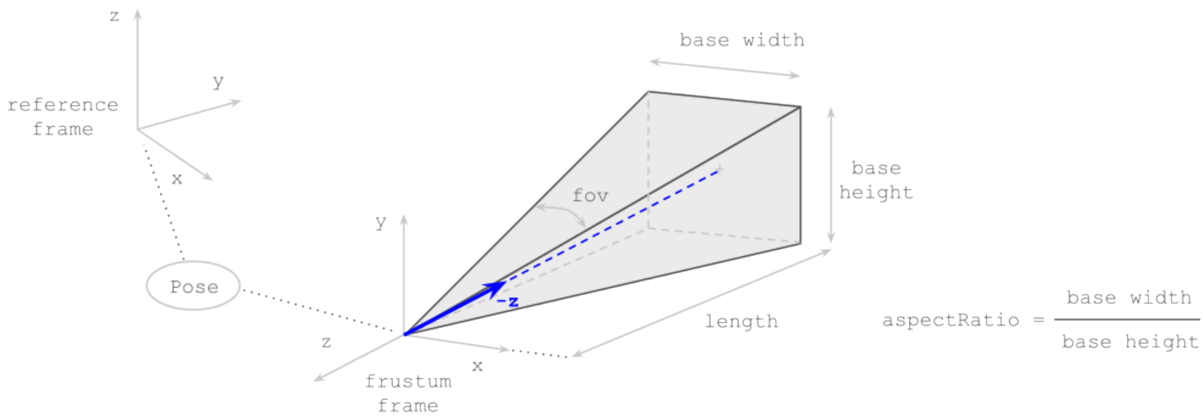

Figure 2 — Pyramidal Viewing Frustum (along -Z)

The shape and parameters of a conical frustum are illustrated below:

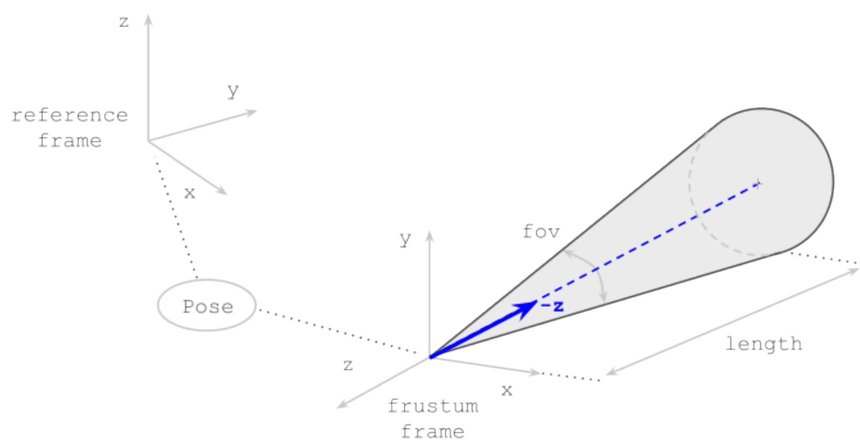

Figure 3 — Conical Viewing Frustum (along -Z)

5.11. Requirements Class "Spherical Sector"

5.11.1. Overview

REQUIREMENTS CLASS 10	
IDENTIFIER	`/req/spherical-sector`
TARGET TYPE	Web API
PREREQUISITES	/req/sampling-features Requirements class 5: `/req/sampling-parametric`
NORMATIVE STATEMENTS	Requirement 22: `/req/spherical-sector/type` Requirement 23: `/req/spherical-sector/attributes`

A `Spherical Sectors` is a 3D sampling feature whose geometry is defined as a part of a sphere. `Spherical Sectors` are often used to model the area that is viewable by a remote sensor or reachable by an actuator.

Examples of Spherical Sectors

- The entire area viewable by a PTZ camera
- The entire area viewable by a radar
- The geographic area reachable by a weapon system

5.11.2. Feature Type

REQUIREMENT 22	
IDENTIFIER	`/req/spherical-sector/type`
INCLUDED IN	Requirements class 10: `/req/spherical-sector`
STATEMENT	The `Feature Type` attribute of a `Spherical Sector` resource SHALL be set to: • The URI http://www.opengis.net/def/samplingFeatureType/x-CS-TBD/1.0/SphericalSector, or • The CURIE `cs:SphericalSector`.

REQUIREMENT 22

	If using a URI or CURIE is not practical, the encoding SHALL unambiguously map one of the possible values used to encode feature types to this URI.

5.11.3. Attributes

REQUIREMENT 23

IDENTIFIER	`/req/spherical-sector/attributes`
INCLUDED IN	Requirements class 10: `/req/spherical-sector`
A	A `Spherical Sector` feature SHALL support the attributes listed in Table 10.
B	The value assigned to these attributes SHALL conform to the specified definition and data type.
C	All attributes marked as `Required` SHALL be present on an instance of the class.

Table 10 — Spherical Sector Attributes

NAME	DEFINITION	DATA TYPE	USAGE
`radius`	Radius of the spherical sector, in meters.	Number	Required
`innerRadius`	Inner radius of the spherical sector, in meters (defaults 0).	Number	Optional
`minAzimuth`	Start azimuth of the spherical sector, in degrees (defaults to 0).	Number	Optional
`maxAzimuth`	Start azimuth of the spherical sector, in degrees (defaults to 360).	Number	Optional
`minElevation`	Start elevation of the spherical sector, in degrees (defaults to -90).	Number	Optional
`maxElevation`	End elevation of the spherical sector, in degrees (defaults to +90).	Number	Optional

The shape and parameters of a spherical sector are illustrated below:

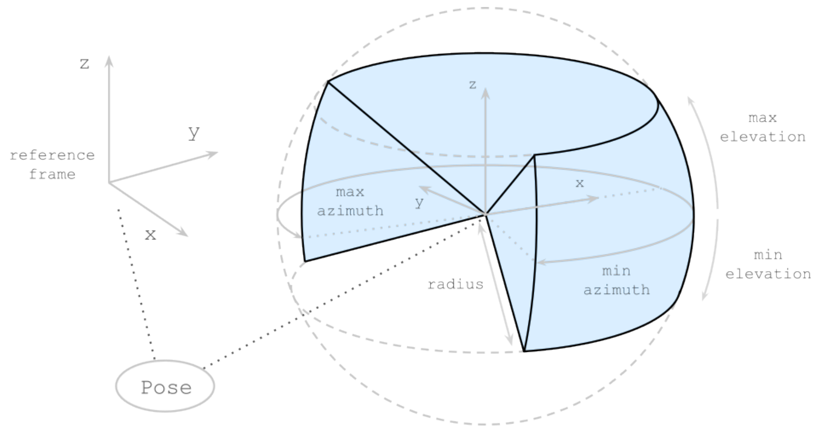

Figure 4 — Spherical Sector

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REQUIREMENTS CLASSES FOR ENCODINGS

6 REQUIREMENTS CLASSES FOR ENCODINGS

This working-draft branch preserves the encodings anchor referenced by the integrated sampling-feature-types clause. Part 4-specific encoding requirements have not yet been promoted into this standalone draft from source material available in this branch.